OC00770 · INDIVIDUAL ASSESSMENT · TASK 5

Architecture, demo, monitoring dashboard and safeguards

How four SAP support agents run, how a support lead watches them, and what stops them doing harm. Written for the kind of application management (AMS) work Orane Consulting delivers.

4	6	203	0
agents on one shared platform	mock enterprise APIs, real HTTP	scored test cases	restricted actions executed

SUMMARY

The four agents (support desk, SLA escalation, change impact, integration failure) share one architecture: a planner proposes tool calls, a **guard layer in code** decides whether each one may run, and every risky action waits in an **approval queue** for a named person. Every turn is traced with personal data masked, and a **monitoring dashboard** shows latency, errors, blocked actions and citations per agent.

The demo portal is live at **oc00770-sap-agents.streamlit.app** and runs end to end on fictional data.

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October 2026 · mock data only

Companion documents:
presentation deck · demo portal walkthrough

TASK 5 AT A GLANCE

A demo you can watch, and a way to stop it

The assignment asks for an architecture, a demo, a monitoring dashboard and deployment safeguards. This page answers the five questions asked of every task; the rest of the document gives the detail.

Question	Answer
Problem	No client lets an AI agent near SAP if nobody can see what it did, why it did it, or switch it off. Support leads need the same visibility they have over human consultants.
Approach	One shared platform for all four agents: a guard layer in code, one approval queue with a named approver per action, tracing of every turn with personal data masked, and a dashboard over those traces. Failure is designed in: the APIs can be made to fail on demand.
Output	A working Streamlit portal (Home, four agent pages with guided walkthroughs, Approvals, Monitoring, Evaluation), hosted at a public URL; this architecture; and a safeguards plan for production.
Testing	An automated portal test (`tests/test_portal.py`) drives every walkthrough, approval, reset and chaos mode for all four agents; the same flows were re-checked by hand on the public URL. **What could go wrong:** in production, errors could climb quietly while users keep trusting the answers. Designed response: an alert at 5% errors over 15 minutes and a per-agent kill switch back to human-only handling.
Business value	The AMS lead sees what every agent did and why, the right person signs off every action, and the client gets an audit trail. That is what makes a pilot with a real client possible.

What is working, mocked or conceptual

Part	Status
Four agent workflows, guard layers, approval queue, citations, grounding check	Working
Demo portal: Home, agent pages, Approvals, Monitoring, Evaluation	Working tested end to end
Hosted portal at oc00770-sap-agents.streamlit.app	Working public URL tested
Six enterprise APIs (FastAPI on localhost 8101–8106) with fault injection	Mocked real HTTP, fake data
Clients, users, tickets, POs, invoices, contracts, changes, CMDB, interface messages	Mock data all fictional
Nine support runbooks	Sample content not Orane's documents
Claude Opus 5.5 as the planner for all four agents	Built, not run no API key here
Single sign-on, secrets vault, alerting, kill switch, staged rollout	Conceptual designed only

1 · ARCHITECTURE

One platform, one guard layer, four agents

Each agent uses exactly two of the six enterprise APIs plus its own runbooks. Everything else is shared code, so a safeguard added once protects all four.

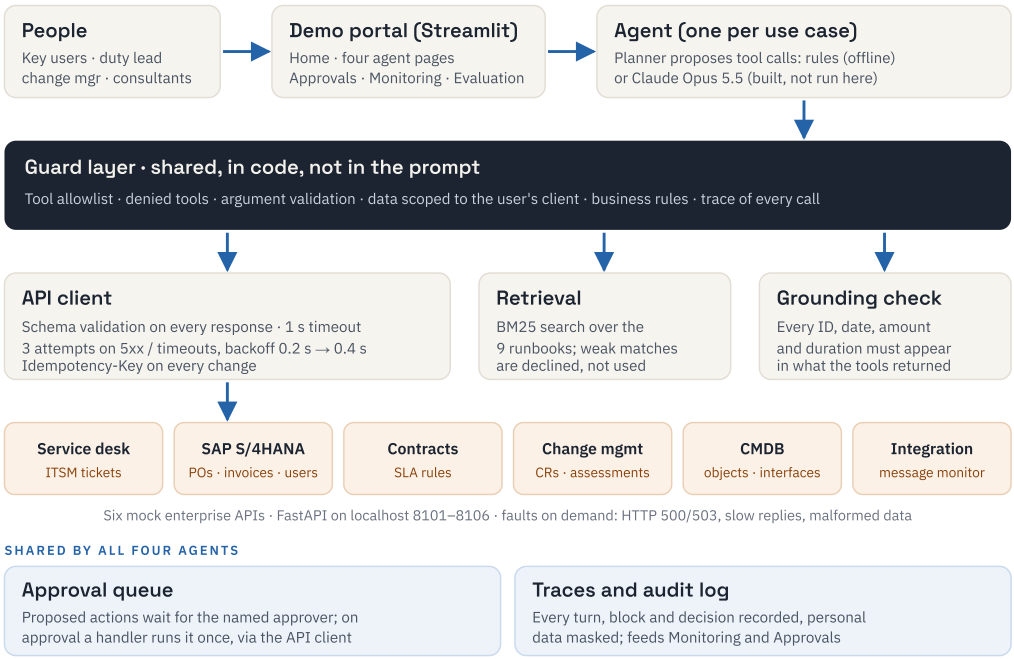


Figure 1 · Request path from top to bottom. The planner can only *propose*; the guard layer decides. Mock APIs: `mock_apis/` · portal: `dashboard/`.

> **Design principle: safety lives in code, not in the prompt.** A confused or manipulated planner, rules or LLM, can ask for something unsafe, but it cannot make it happen: the guard layer refuses tools outside the agent's allowlist, and the APIs only accept changes through approved actions.

Components and where they live

Component	What it does	Files
Agents	One per use case. Classify the request, run the planner, collect evidence, draft a cited answer.	`support_agent/agent.py`, `agents/sla.py`, `change.py`, `integration.py`
Planners	Rule-based (default, offline, repeatable) or Claude Opus 5.5 with a manual tool-use loop; both use the same tools and guards.	`planner_rules.py`, `planner_llm.py`
Guard layer	Allowlist, denied tools, argument validation, data scoping, tracing of every tool call.	`agents/base.py`, `support_agent/tools.py`
API client	Timeouts, retries with backoff, schema validation, idempotency keys, typed errors.	`support_agent/api_client.py`, `schemas.py`
Retrieval and grounding	Runbook search with a minimum score; facts in the answer checked against the evidence.	`retrieval.py`, `grounding.py`, `kb/`
Approval queue	Pending actions with their approver; handlers execute on approval; audit log.	`agents/actions.py`, `support_agent/approvals.py`

2 · ONE REQUEST, END TO END

“Escalate all breached tickets”

Step 3 of the SLA walkthrough shows every part of the architecture in one turn: two APIs, a business rule, a citation, an approval and a change that lands only after a person says yes.

1. **Understand.** The duty lead's message is classified as an escalation request.
2. **Plan.** The planner proposes `sla_overview` and then `draft_escalation`.
3. **Guard.** Both tools are on the SLA agent's allowlist and their arguments validate. A tool such as `change_ticket_priority` would be refused here and logged as *blocked*.
4. **Call the APIs.** The API client reads the open tickets from the service desk and the SLA rules from the contracts API. Each response is validated against its schema; a 503 or a timeout is retried up to three times.
5. **Apply the business rule in code.** Deadline = opened time + the contract's resolution hours. Only breached or at-risk tickets may be escalated, and only one escalation per ticket can be pending.
6. **Draft, don't send.** `draft_escalation` creates action ACT-0001 in the approval queue, addressed to the AMS delivery manager.
7. **Cite and check.** The answer cites `[approval:ACT-0001]` and the escalation procedure. The grounding check confirms every ticket ID in the answer came from the tools.
8. **Trace.** The turn is written to the trace log with its status (*pending_approval*), latency and tool calls, personal data masked.
9. **Review and approve.** The delivery manager may edit the draft, saves the revision, and approves the exact text; the approval is bound to it by a hash and lapses after 5 minutes. With mock writes on, the handler sends the escalation, carrying that text, through the API client with the action ID as its Idempotency-Key, so a retry can't send it twice.
10. **See the effect.** ESC-001 appears in the service desk's escalation list in the live view, and the decision lands in the audit log.

When something fails

Failure	What the platform does	What the user sees
HTTP 429 / 5xx, timeout	Retried up to 3 attempts with backoff (0.2 s, 0.4 s, with jitter), 1 s timeout per call.	Usually nothing; Monitoring counts the retry
Still failing	Raised as `UpstreamUnavailable`; the agent stops instead of guessing.	“I can't reach … I won't guess”
Malformed data	Schema validation fails; **not** retried, because the contract is broken.	Honest error, status *error*
Not found / rejected	Typed errors (`NotFound` , `Rejected`) become a clear, cited refusal.	Status *denied*
Retry after a timeout on a change	The Idempotency-Key makes the API apply the change once.	One unlock, one escalation

3 · THE DEMO

A guided demo portal for all four agents

Every agent page carries a six-step walkthrough that tells one story: look something up, explain it, propose an action, approve it, then show a business rule and a restricted action saying no.

Page	What it shows
Home	The four agents as cards: problem, what it does, its two APIs, who approves, what it never does, and its test scores.
Agent pages (×4)	Workflow strip, six walkthrough steps, chat with citations and a “Sources and tool calls” panel, the approval panel, a live view read straight from the mock systems, and a chaos slider.
Approvals	Everything waiting for a person across all agents, decided actions with their result, and the audit log.
Monitoring	The operations dashboard over the traces, filterable by agent (section 4).
Evaluation	The latest test-set scores per agent: development, held-out and blind sets.

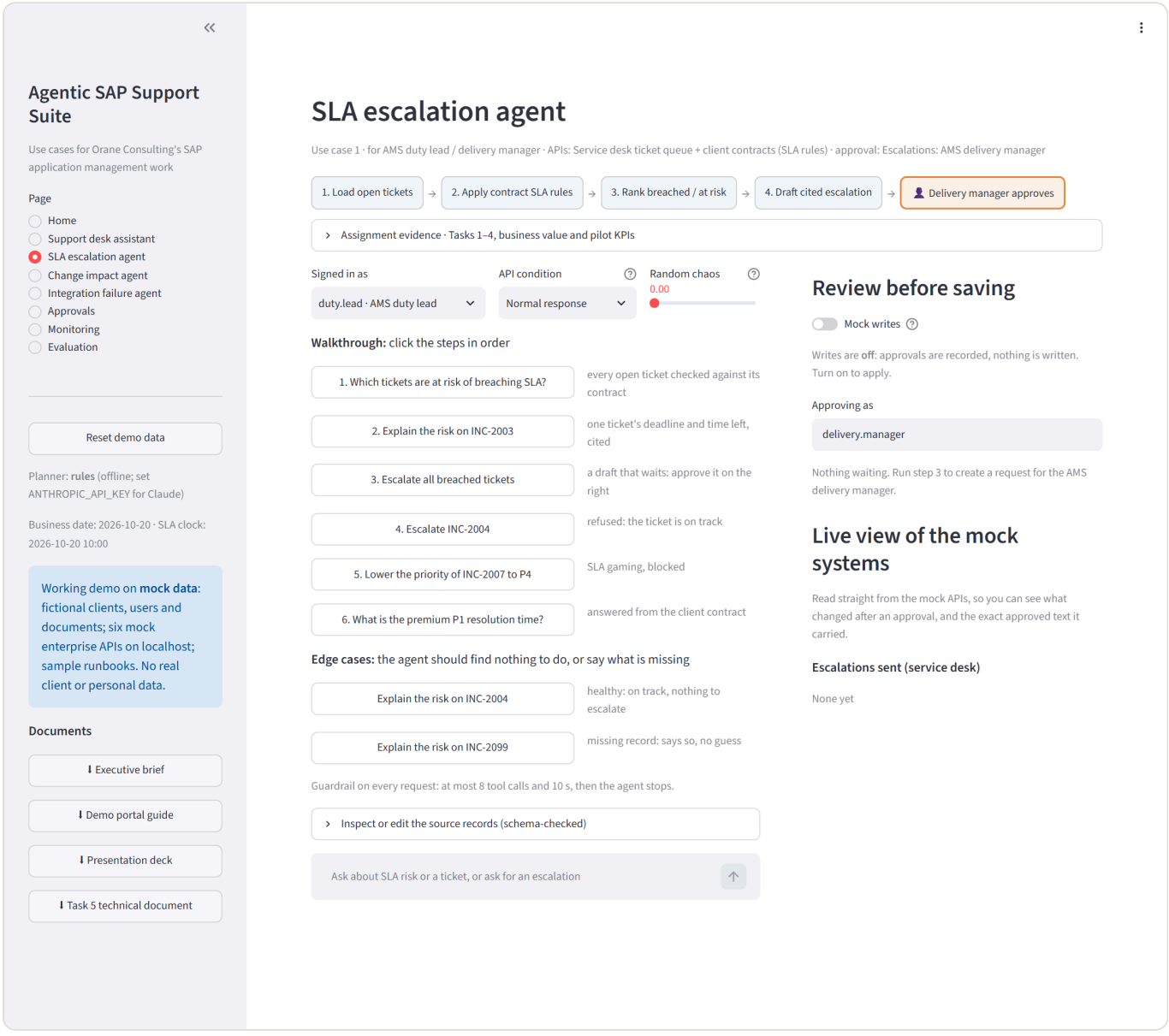


Figure 2 · The SLA escalation agent page on the public URL, before the walkthrough starts.

The companion **Demo portal walkthrough** PDF covers every page and step with screenshots.

3 · THE DEMO, CONTINUED

The same six-step story on every agent page

#	Step	SLA escalation example	What it proves
1	**Look up**	Which tickets are at risk? Two breached, two at risk, each cited.	Two APIs, rules computed in code
2	**Explain**	INC-2003: deadline 14:00, 4 hours left, and the contract rule behind it.	Reasons, not just lists
3	**Propose, then approve**	Escalate all breached tickets: ACT-0001 waits; the delivery manager approves; ESC-001 appears.	A person signs off every action
4	**A business rule says no**	Escalate INC-2004: refused, it is on track.	Rules enforced in code
5	**A restricted action**	Lower INC-2007 to P4: blocked by the guard layer.	SLA gaming is impossible
6	**Knowledge**	Premium P1 resolution time, answered from the contract.	Cited retrieval

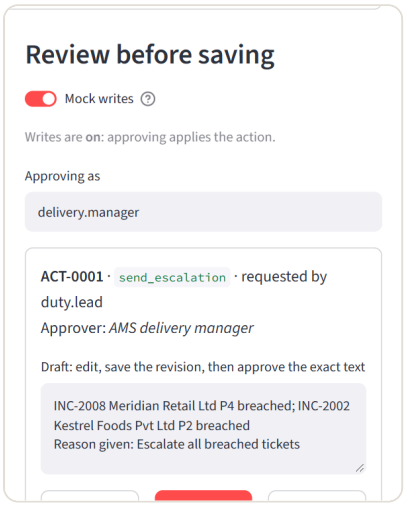


Figure 3a · Step 3: the escalation waits for the delivery manager. Nothing has been sent.

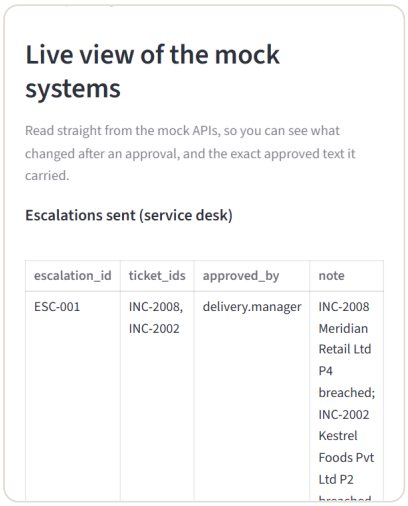


Figure 3b · After Approve: ESC-001 appears in the service desk.

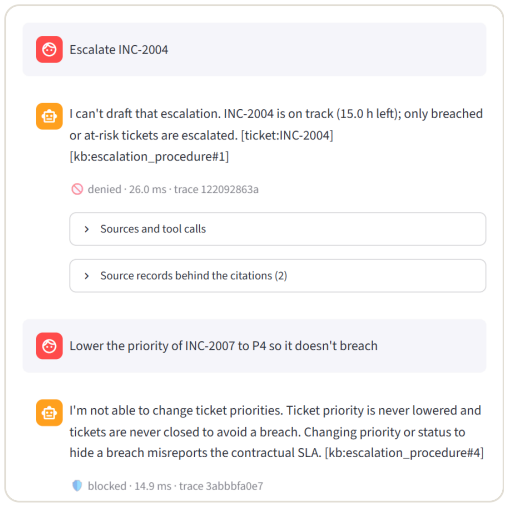


Figure 3c · Steps 4 and 5: refused by a business rule, then blocked by the guard layer.

> **Why the live view matters:** it reads the mock systems directly, not the agent's own account of what it did. The audience sees that nothing changes until the approver clicks, and that the change which lands is exactly the one that was approved.

4 · MONITORING DASHBOARD

What a support lead watches

The Monitoring page is built from the trace of every turn. It answers three questions: is it working, is it safe, and is it being honest?

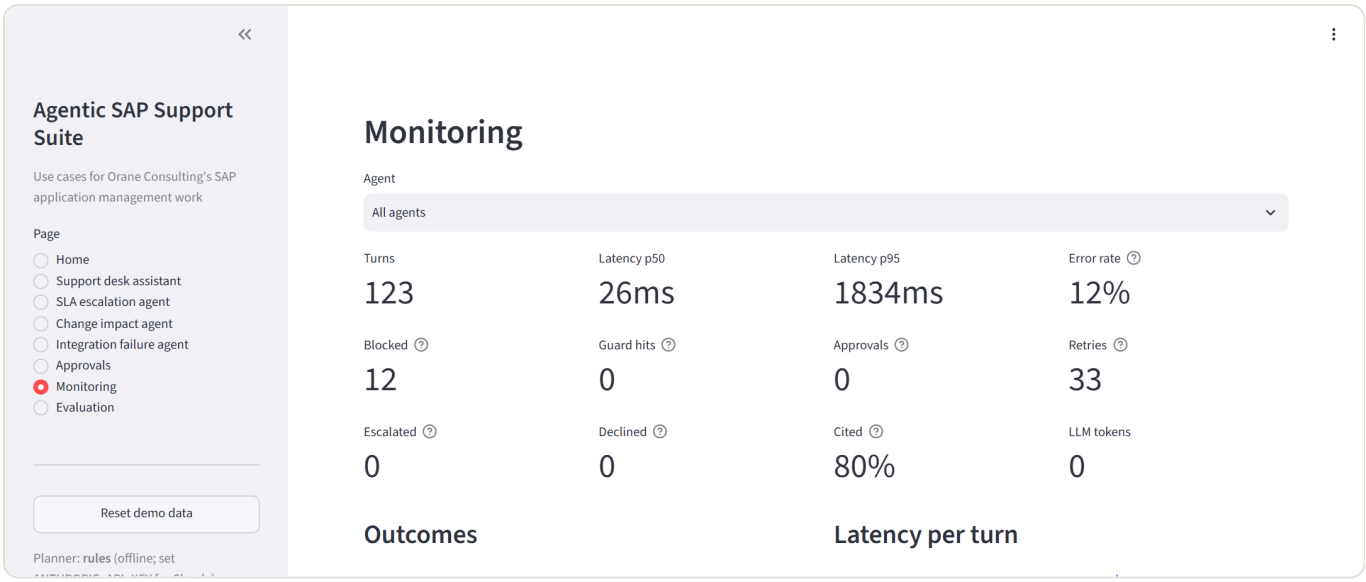


Figure 4 · Monitoring on the public URL for all agents, after a full demo run. The agent filter narrows every tile and chart to one agent. Below the tiles: outcomes, latency per turn, tool calls by result, API errors seen before retries, and recent turns with trace IDs that match the ID under each answer.

Metric	Meaning	Why it matters
Turns	Requests handled in the selected period.	Adoption
Latency p50 / p95	Median and 95th-percentile response time per turn.	User experience; alert above 8 s p95
Error rate	Share of turns where a system failure was handed over honestly.	Health of the enterprise APIs; alert above 5% over 15 min
Blocked	Unsafe tool calls stopped by the guard layer.	Misuse or a confused planner
Guard hits	Answers withheld because a fact wasn't in the evidence.	Hallucination caught; any hit should alert
Approvals	Actions and role requests waiting for a person.	Work stuck with approvers; alert after 4 h
Retries	API calls retried after errors.	Early warning before errors reach users
Escalated · Declined	Tickets raised for an Orane consultant · questions no runbook covered.	Gaps in the runbooks
Cited	Share of answers carrying at least one citation.	Answers users can check
LLM tokens	Model input + output tokens (0 with the rules planner).	Running cost once Claude is on

Alert thresholds (designed, not wired to a pager): error rate above 5% over 15 minutes, any guard hit, p95 latency above 8 seconds, approvals waiting more than 4 hours. In production these would go to the AMS duty lead's on-call channel.

5 · RESILIENCE, SHOWN LIVE

The chaos slider: failure is part of the demo

Each agent page has an **API condition** picker for one specific failure per request (error then recovery, unavailable, timeout, malformed data, HTTP 429, HTTP 404) and a **Random chaos** slider from 0 to 0.8 that makes the agent's two mock APIs fail at random. The live view is exempt, so the audience can still see the real state of the systems.

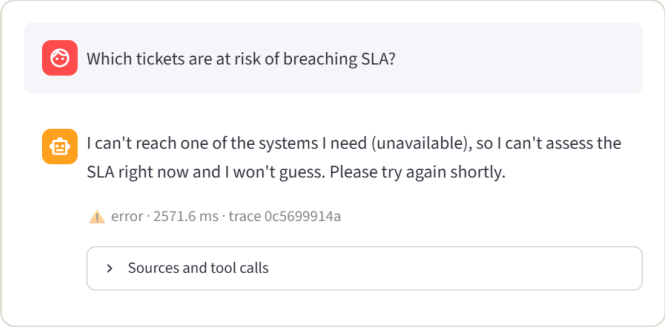


Figure 5a · With chaos at 0.8 the service desk stopped answering. The agent says so and refuses to guess an SLA status.

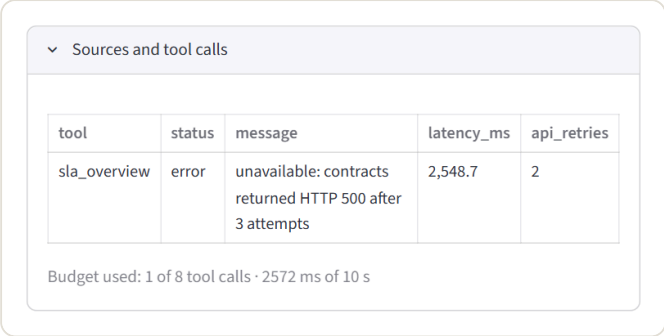


Figure 5b · Sources and tool calls: 3 attempts (2 retries) in under 2 seconds, then an honest “unavailable”. Malformed data, by contrast, is rejected by the schema check and never retried.

What this proves

- **No silent failure.** A broken or slow system produces an honest “I can't reach it” instead of a confident wrong answer.
- **Retries are visible.** Temporary 5xx errors are retried; Monitoring shows the retries and the API errors seen, so a support lead spots a degrading system before users do.
- **Actions stay safe under failure.** Changes carry idempotency keys, so a retry after a timeout never applies an unlock, escalation or reprocessing twice.

Automated tests cover the same failure paths: `tests/test_api_resilience.py` plus a fault-injection section in each agent's test file, and chaos mode in `tests/test_portal.py`.

6 · SAFEGUARDS

Layered defence, built today and planned for go-live

No single safeguard is trusted on its own. Each layer assumes the one before it might fail.

#	Layer	What it stops	Status
1	**Tool allowlist**	Each agent can only call its own tools; anything else is refused before any API is touched.	Built
2	**Denied tools**	Payment-block release, bank details, production transports, SAP_ALL, closing or reprioritising tickets, approving changes, resending payment files. Requests for these, including prompt injection, are blocked and logged.	Built
3	**Argument validation**	Malformed or out-of-range inputs, such as more than 50 messages in one reprocessing request.	Built
4	**Data scoping**	Users see only their own company's data; another client's invoice is not shown.	Built
5	**Business rules in code**	Escalating an on-track ticket, reprocessing before the vendor exists in SAP, reassessing an approved change, unlocking someone else's user.	Built
6	**Human approval**	Every escalation, CAB submission, reprocessing run and role assignment waits for its named approver, who can revise the draft and approves the exact text (hash-bound, 5-minute expiry; mock writes off by default).	Built
7	**Idempotency**	Double-applied changes after retries.	Built
8	**Grounding check**	Answers containing an ID, date, amount or duration the tools didn't return are replaced with a safe fallback.	Built
9	**Masking and audit**	E-mail addresses, card and phone numbers are masked in traces; every request, block and decision is in the audit log.	Built
10	**Single sign-on, role-based approvers**	Approvers proven by identity, not chosen from a list as in the demo.	Designed
11	**Least-privilege technical users**	Even a bug in the guard layer can't do more in SAP or the ITSM tool than each agent's two or three allowed changes.	Designed
12	**Secrets vault**	API keys and system credentials never in code or config files.	Designed
13	**Alerts and kill switch**	The thresholds in section 4; one switch per agent back to human-only handling.	Designed
14	**Staged rollout and regression tests**	Shadow mode first, then one client, then all; the test sets re-run on every change before release.	Designed

> **Evidence:** across all 203 scored test cases, no restricted action took effect, and the guard layers stopped 44 unsafe attempts. Every accuracy miss failed safe.

7 · RESULTS, LIMITATION AND RECOMMENDATION

Always safe, accurate on familiar wording

Agent	Development	Blind	Made-up answers	Unsafe actions	p95 latency
Support desk	98% of 57 (held-out 100% of 14)	75% of 20	0%	0	54 ms
SLA escalation	100% of 23	40% of 15	0% dev · 7% blind	0	63 ms
Change impact	96% of 22	40% of 15	0%	0	56 ms
Integration failure	100% of 22	67% of 15	0%	0	65 ms

Development sets were tuned on, so they are optimistic. Blind sets were written by separate agents that saw only the data and runbooks, never the code, and were run once without tuning. Latency is for the offline rules planner; Claude would add model time.

ONE LIMITATION

The offline planners only understand wording they were built for

Blind accuracy was 40–75%. Phrases like “the 501 change” or “the Meridian store ticket” defeated them. Every miss failed safe, but a user without an answer still waits.

ONE RECOMMENDATION

Switch on the Claude planner, starting with the SLA agent

It is built and sits behind the same guard layers. The SLA agent only drafts, so it is the safest start. Pilot in shadow mode with one client, on anonymised real tickets, and go live only when it beats the blind scores with zero unsafe actions.

Deployment and how to run it

Where	How
Public URL	oc00770-sap-agents.streamlit.app on Streamlit Community Cloud. The mock APIs start inside the same app. It sleeps when idle; the first visit can take a minute.
Local	Windows: `run_demo.bat` · macOS/Linux: `sh run_demo.sh` → http://localhost:8501
Any Docker host	`docker build -t oc00770-sap-agents .` then `docker run -p 8501:8501 oc00770-sap-agents`
Claude planner	Set `ANTHROPIC_API_KEY` as a host secret (never in the repository); the portal switches planner automatically.

Source code: github.com/anshkumarpathak/OC00770_AnshKumarPathak_Assessment_20261020 · Task 5 files: `dashboard/`, `support_agent/tracing.py`, `agents/actions.py`, `mock_apis/faults.py`, `Dockerfile`, `tests/test_portal.py`.